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A review of the use of finite element simulation and machine learning techniques in the morphological analysis of breast tissue

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ABSTRACT

With the advancement of modern medicine and engineering, breast deformation analysis has become increasingly important in the fields such as cosmetic surgery, reconstructive surgery, sports biomechanics, and health-related research. The morphological changes in breast tissue not only affect an individual's aesthetic appearance and mental well-being but are also closely linked to breast health. Finite Element Method (FEM) and Machine Learning (ML) are two advanced technologies that exhibit significant potential in breast deformation analysis. FEM plays a crucial role in modeling the mechanical behavior of breast soft tissues and analyzing deformations due to its precise numerical solutions and realistic simulation capabilities. On the other hand, machine learning techniques provide new perspectives for personalized health management and disease risk assessment by processing large-scale breast morphology data to uncover patterns and correlations. This paper reviews the latest advancements in the application of FEM and ML in breast tissue morphology analysis, explores their potential and challenges in simulating both static and dynamic breast deformations in clinical practice, and summarizes the characteristics and application scenarios of both technologies. This paper discusses the novel opportunities brought by the integration of these two approaches for the clinical diagnosis and analysis of breast-related diseases. It synthesizes these developments and explores the potential benefits of methodological integration for future breast morphology-related research and clinical applications.

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1. Introduction

With the advancement of medicine and engineering, the morphological analysis of breast tissue has become increasingly important in the fields such as plastic and reconstructive surgery, sports biomechanics, and health-related research (Zhang et al. 2022). As a prominent feature of the female body, changes in breast morphology not only affect an individual's aesthetic appearance and mental well-being but are also closely linked to breast health (Teixeira and Martins 2023). Abnormal deformations caused by breast movement may lead to discomfort and pain (Gilmer et al. 2024), as well as abnormal hyperplasia of breast tissue, which in turn increases the risk of diseases such as breast cancer. In recent years, finite element method (FEM) and machine learning (ML) have emerged as two advanced technologies that demonstrate significant application potential in the field of breast deformation analysis.

With its numerical solving capabilities and realistic simulation effects, the finite element method is widely used to model breast tissues for analyzing mechanical behavior (Mira et al. 2018) and deformation (del Palomar et al. 2008; Danch-Wierzchowska et al. 2020) during motion. This method allows for detailed analysis of the mechanical responses of breast tissue models by partitioning them into multiple small, manageable elements, thereby facilitating the examination of various external conditions. The finite element method not only simulates the dynamic behavior of the breast during movement, such as running (Chen et al. 2024) or other physical activities (Liu et al. 2023), but also accurately models the dynamic contact interactions between the breast and undergarments (Liang et al. 2019; Sun et al. 2019; 2021; Liu et al. 2023). This method aids in the design of sports bras and activewear that meet physiological needs, evaluating the performance *via* simulated biomechanical data thus reducing excessive

movement and discomfort of breast tissue during physical activity. Furthermore, it can model the compressive deformation of the breast during medical examinations (Chung et al. 2008; Han et al. 2012; Sturgeon et al. 2016; Mîra et al. 2018) and the deformation under various positions (del Palomar et al. 2008; Gamage et al. 2012; Chen et al. 2019; Mazier and Bordas 2024), which helps optimize imaging quality and improve diagnostic accuracy, guiding clinical practice in examination strategies (Ringel et al. 2023). For example, the finite element method can predict morphological changes in breast tissue following cosmetic surgery (Roose et al. 2005; Thanoon et al. 2015), assisting in the optimization of surgical plans to improve postoperative outcomes and minimize discomfort (Haddad et al. 2016). It is also utilized in the design and optimization of breast reconstruction materials by simulating the mechanical properties of different reconstruction materials (Dong et al. 2016; Myung et al. 2019), including the morphology (Roose et al. 2005; Dong et al. 2016), function (Myung et al. 2019), and biomechanical responses of the breast post-reconstruction (Thanoon et al. 2015; Dong et al. 2016; Haddad et al. 2016; Myung et al. 2019). This facilitates the selection of the best reconstruction strategies in breast surgery, enhancing patient comfort and recovery quality.

Meanwhile, machine learning technologies, by processing large-scale breast morphology data (Martínez-Martínez et al. 2017; Wang et al. 2019; Zhang et al. 2022), can uncover implicit patterns and associations within the data, providing new perspectives and solutions for personalized health management and disease risk assessment. Decision tree (Wels et al. 2012; Martínez-Martínez et al. 2017; Zolfagharnasab et al. 2018; O'Neill et al. 2020) algorithms facilitate the classification and regression analysis of data, aiding in the identification of various breast morphology features and their potential health risks, thereby supporting preliminary risk assessment and screening recommendations. Neural networks (Ganggayah et al. 2019; Said et al. 2022; Zhang et al. 2022; Feng and Samani 2023), particularly deep neural networks, leverage multi-layered nonlinear transformations to extract intricate features from complex breast imaging data, enhancing the precision of breast tissue condition identification and classification. Deep learning models (Mendizabal et al. 2020; Chugh et al. 2021; Pfof et al. 2022), as advanced forms of neural networks, further improve the predictive capabilities regarding changes in breast morphology. These models enable automated complex image analysis and pattern recognition, supporting efficient early breast cancer screening and offering personalized health management strategies. By applying

these technologies to the analysis of breast morphology data, machine learning not only reveals hidden disease patterns but also enhances the accuracy of early detection of breast diseases and optimizes individualized treatment plans, significantly improving patient health management and treatment outcomes.

This paper reviews the latest advancements and applications of finite element simulation and machine learning methods in the analysis of breast deformation in recent years. It explores the capability of finite element method to simulate breast deformation under various conditions, as well as its potential and challenges in clinical practice. Additionally, it summarizes the research progress and development trends of machine learning in breast deformation studies, comparing the characteristics and application scenarios of the two technologies. The aim is to provide theoretical support and practical guidance to further promote the development and application of breast morphology analysis techniques.

2. Application of the finite element method in breast deformation analysis

2.1. Establishment of breast finite element models

2.1.1. Geometric reconstruction

In breast deformation analysis, geometric reconstruction is a critical step aimed at obtaining accurate three-dimensional geometric information of the breast, providing a foundation for subsequent finite element model development and simulation analysis. In addition to utilizing medical imaging techniques such as MRI (Chung et al. 2008; Gamage et al. 2012; Han et al. 2012; Eder et al. 2014; Haddad et al. 2016; Mîra et al. 2018; Chen et al. 2019; Danch-Wierzchowska et al. 2020; Ringel et al. 2023; Mazier and Bordas 2024) or CT (del Palomar et al. 2008; Hsu et al. 2011; Sturgeon et al. 2016; Fedon et al. 2021) for geometric reconstruction, body scanning technologies can also be employed to obtain point cloud data of the human body, which can then be converted into geometric models using reverse engineering software (Chen et al. 2024; Liu et al. 2023; Sun et al. 2019; Sun et al. 2021; Liang et al. 2019; Zhang et al. 2022).

Prior to geometric reconstruction using medical imaging data such as MRI or CT, a standardized pre-processing pipeline is typically required. This generally includes image denoising to improve the signal-to-noise ratio, intensity normalization to enhance comparability across different scans, and registration of multi-sequence or multi-phase images to ensure precise spatial alignment of anatomical structures.

Subsequently, through manual, semi-automatic, or fully automatic segmentation algorithms, the processed image voxels are classified into specific tissue types, such as skin, fat, glandular tissue, muscle, and possible lesions. This segmentation result directly defines the geometric boundaries and spatial distribution of different material regions in the finite element model, representing a critical step in transforming discrete imaging data into a continuous computational model. Hsu et al. (2011) developed a semi-automatic segmentation algorithm for a specialized breast CT dataset to create a compressible breast model. This algorithm segmented the denoised and scatter-corrected dataset into three components: adipose tissue, glandular tissue, and skin. They generated a mesh composed of four-node tetrahedral elements and utilized an explicit finite element solver to simulate breast compression, analyzing several properties to evaluate their impact on the compressible model. Fedon et al. (2021) utilized patient images from a specialized breast CT with isotropic spatial resolution, along with a series of image processing algorithms, to classify each voxel in the reconstructed breast CT images into four categories: air, skin, adipose tissue, and fibro-glandular tissue. By leveraging the intensity and regional information within the images, they converted the classified breast CT images into a finite element biomechanical model to simulate breast compression during mammographic examinations or digital breast tomosynthesis (DBT) procedures. Eder et al. (2014) performed semi-automatic segmentation of different relevant anatomical regions of interest from an MRI dataset, representing the breast soft tissue as components of glandular tissue, adipose tissue, muscle, and skin. They varied the material properties to analyze and predict breast deformation. Mira et al. (2018) developed and evaluated a new biomechanical finite element model that incorporates the stress-free breast geometry and constitutive models derived from tissue deformation measurements in MRI images, including complex anatomical structures such as adipose tissue, glandular tissue, muscle, skin, suspensory ligaments, and fascia. Mazier and Bordas (2024) segmented MRI images into breast tissue, adipose tissue, pectoralis major muscle, skin, and tumors, establishing subject-specific geometric models and proposing a simulation pipeline that allows for the computation of patient-specific geometries and the prediction of personalized breast material properties.

Body scanning technology focuses on capturing surface geometric data, making it suitable for large-scale sample collection and overall morphological analysis,

though it cannot provide detailed information about internal structures. This data requires segmentation using reverse engineering software to construct geometric models. Liu et al. (2023) employed a non-contact three-dimensional scanner to acquire point cloud data from study participants' entire bodies, importing this data into the reverse engineering software Geomagic Wrap to create non-uniform B-spline curves for the upper body, sports bra, and padding. Through processes of trimming, repairing, and smoothing, they developed a NURBS surface model for the upper body, where the finite element geometric model comprises a pair of soft breasts and a rigid upper body. Zhang et al. (2022) proposed a three-dimensional nonlinear biomechanical finite element model that considers complex anatomical structures, including soft tissues (torso, breasts, pectoralis major) and hard bones (humerus, sternum, clavicle, ribs), to simulate breast injuries during lateral arm elevation in the elderly. Chen et al. (2024) developed a novel nonlinear multi-component dynamic finite element method that utilized experimental data from 4D scanning technology and motion capture systems to construct a geometric model, incorporating rigid bodies, soft tissue layers, skin, pectoralis major, adipose tissue, ligaments, and glandular tissue.

2.1.2. Mesh generation

Using the finite element method, the geometric shape of the breast is converted into a discrete finite element mesh. These meshes are typically divided into numerous finite element units, such as tetrahedra or hexahedra, to facilitate the numerical analysis of the mechanical behavior within the breast. Generally, a finer mesh leads to easier convergence of calculations and more accurate results; however, this also increases computational time, costs, and complexity. Therefore, selecting the most appropriate element type and controlling the mesh density is crucial for constructing a finite element model (McCurry et al. 2015).

In the finite element modeling of breast deformation, mesh generation must consider both the type of mesh elements and the mesh size. Azar et al. (2001) divided the breast model into soft tissue and skin layers, employing linear eight-node hexahedral solid elements for the 3D breast mesh and modeling the skin with three-node triangular shell elements. To predict the location of tumors within the glandular tissue, they refined the mesh around tumor points in the MR images, with mesh density decreasing with distance from the refinement area. Samani et al. (2001) established a breast model using eight-node hexahedral elements and four-node skin membrane elements,

comparing two meshing techniques: voxel-based (VB) and tetrahedral interpolation (TI). The results indicated that the TI mesh more accurately represented the breast shape, yielding superior simulation outcomes.

Due to the complex geometry, generating high-quality hexahedral meshes is challenging (Bourdin et al. 2007). del Palomar et al. (2008), Han et al. (2010), and Chen et al. (2013) utilized linear four-node tetrahedral elements for meshing breast tissue. The breast tissue is typically divided into tetrahedral elements of sizes such as 4 mm (Roose et al. 2005), 5 mm (Azar et al. 2000; Thanoon et al. 2015; Sun et al. 2019; 2021), 6 mm (Liang et al. 2019), and 8 mm (Liu et al. 2023). Liang et al. (2019) constructed models with five different mesh sizes (2, 4, 6, 8, and 10 mm) to assess mesh dependency in the finite element model. They found that the dynamic breast displacement differed by only 0.26% between the 4 mm and 6 mm mesh sizes. However, the computational time for the 4 mm model was three times longer than that for the 6 mm model. Considering computational efficiency and accuracy, they ultimately chose the 6 mm mesh size for subsequent analyses.

Tanner et al. (2006) utilized quadratic ten-node tetrahedral elements to mesh breast tissue, testing various element sizes ranging from 7.5 mm to 10 mm. Their results indicated that the model's accuracy was largely unaffected by changes in element size. Hsu et al. (2011) et al. investigated the application of first-order and second-order elements in breast deformation simulations. The corner nodes of the quadratic ten-node tetrahedral elements were able to withstand bending loads, providing more accurate results. The study found that, compared to linear tetrahedral elements, quadratic tetrahedra generally yielded more precise simulations at higher mesh densities. By testing element lengths of 1.5 mm, 2.5 mm, 3.75 mm, 5 mm, and 1 cm, they observed that larger element sizes resulted in poorer deformation outcomes. The type of element (tetrahedral or hexahedral, linear or quadratic) is crucial in finite element modeling (FEM) and mesh resolution. Tetrahedral elements converge more slowly than hexahedral elements in studies involving incompressible materials (Ruiter et al. 2006). Additionally, modifying the mesh for complex geometries, such as the 3D surface of the breast, can lead to aspect ratio discrepancies and irregular geometries, thereby increasing computational difficulty.

2.1.3. Determination of material properties

In finite element simulation analysis, the definition of materials is a critical process for determining the

physical properties and behavioral characteristics of the model's components. Therefore, selecting an appropriate material model is essential. The mechanical properties of human breast tissue exhibit complex nonlinear behavior, and to ensure the accuracy of the simulations, it is necessary to characterize breast properties using homogeneous and isotropic nonlinear hyperelastic materials. The Mooney-Rivlin (Pianigiani et al. 2015; Liang et al. 2019; Sun et al. 2019; 2021; Zhang et al. 2022) model and the Neo-Hookean (del Palomar et al. 2008; Gamage et al. 2012; Míra et al. 2018; Chen et al. 2019) model are commonly employed hyperelastic materials for characterizing the properties of breast tissue.

del Palomar et al. (2008) explored the spatial correlation between breast characteristics in supine and standing positions, assuming that adipose and glandular tissues are isotropic, hyperelastic, and nearly incompressible. They obtained the corresponding parameters by fitting empirical data to the simplest hyperelastic Neo-Hookean model. Sun et al. (2019) utilized iterative variations of the nonlinear Mooney-Rivlin coefficients to simulate the large deformations exhibited by the breast during forward bending movements, and employing five Mooney-Rivlin (Sun et al. 2019, 2021) coefficients to describe the hyperelastic properties of breast tissue in subsequent studies to analyze the interaction between breasts and bra. Pianigiani et al. (2015) developed a subject-specific finite element model with parameters for healthy breasts, incorporating several nodules to predict large deformations of breast tissue and analyze the relationship between nodule displacement and Mooney-Rivlin material parameters. Han et al. (2012) proposed a novel breast modeling approach using a nonlinear hyperelastic material model, optimizing material model parameters for soft tissues to simulate the interaction between breast tissue and compression plates. The material model parameters were iteratively updated to achieve image similarity between finite element predicted MR images and experimental breast MR images, thereby obtaining optimized deformation predictions.

From a clinical application perspective, the choice between the two commonly used hyperelastic models—Mooney-Rivlin and Neo-Hookean—depends on the simulation objectives, magnitude of deformation, and availability of data. The Neo-Hookean model is concise in form and requires only a few material parameters (typically one or two), relying on a linear relationship between stress and the first invariant of strain. This makes it computationally efficient and sufficiently stable for simulating small to moderate deformations. It is particularly suitable for preliminary analyses where

computational speed is prioritized or material parameters are limited, such as in rapid screening of different surgical plans or large-scale parametric studies. In contrast, the Mooney-Rivlin model (especially in its polynomial form) incorporates additional material constants, enabling a more accurate description of the complex nonlinear mechanical responses of soft tissues under large deformations, including strain-hardening and second-invariant-dependent behavior. Therefore, in scenarios demanding high-fidelity predictions—such as accurately simulating breast dynamics during vigorous exercise (e.g. running, jumping), evaluating complex contact pressure distributions between the breast and implants or bras, and predicting tissue stress states under significant compression to optimize biopsy or imaging procedures—the Mooney-Rivlin model is often the preferred choice. The key to model selection lies in balancing complexity, computational cost, and the required clinical predictive accuracy. The Neo-Hookean model serves as an effective tool for rapid and robust analysis, whereas the Mooney-Rivlin model offers the potential for physically accurate, in-depth simulations.

Table 1 summarizes the material parameter settings for the breast finite element models. Linear elastic models are suitable for cases involving small deformations of glandular tissue, where the relationship between stress and strain can be effectively described by linear models. However, breast tissue typically exhibits complex nonlinear mechanical behavior. Therefore, when large deformations occur or higher simulation accuracy is required, hyperelastic material models are more appropriate. The key factor in selecting a material model lies on the degree of deformation and the precision requirements of the simulation. If the deformation is small and the accuracy requirements are low, a linear elastic material can be chosen, while for larger deformations or higher accuracy demands, a hyperelastic material model should be employed to more accurately simulate the mechanical behavior.

2.2. Analysis of breast deformation under different boundary conditions

Establishing appropriate boundary conditions is a crucial step in the finite element analysis of breast deformation, directly impacting the accuracy and reliability of the simulation results. It is essential to precisely define and apply corresponding boundary conditions based on the actual situations to accurately simulate the biomechanical responses of breast tissues during movement. This involves considering breast motion, interactions with surrounding tissues, and constraints

Table 1. Breast material property settings.

Model Component	Material Property	Material Parameters	References
Skin	Linear Elasticity	$E = 88 \text{ kPa}$, $\nu = 0.49$	Sturgeon et al. (2016)
Gland		$E = 10 \text{ kPa}$, $\nu = 0.49$	
Fat		$E = 1 \text{ kPa}$, $\nu = 0.49$	
Breast	Mooney-Rivlin	$C_{10} = 0.05 \text{ kPa}$, $C_{01} = 0.052 \text{ kPa}$, $C_{11} = 0.375 \text{ kPa}$, $C_{20} = 0.78 \text{ kPa}$, and $C_{02} = 0.63 \text{ kPa}$	Sun et al. (2019) Sun et al. (2021)
Breast	Mooney-Rivlin	$C_{01} = 0.108 \text{ kPa}$, $C_{02} = 1.18 \text{ kPa}$, $C_{10} = 0.094 \text{ kPa}$, $C_{11} = 0.82 \text{ kPa}$, and $C_{20} = 0.84 \text{ kPa}$	Ruixin et al. (2020)
Breast	Mooney-Rivlin	$C_{10} = 0.041 \text{ kPa}$, $C_{01} = 0.0042 \text{ kPa}$, $C_{11} = 0.31 \text{ kPa}$, $C_{20} = 0.65 \text{ kPa}$, and $C_{02} = 0.52 \text{ kPa}$	Liang et al. (2019) Sun et al. (2019)
Skin	Linear Elasticity	$E = 1 \sim 100 \text{ kPa}$, $\nu = 0.49$	Hsu et al. (2011)
Gland		$E = 1 \sim 50 \text{ kPa}$, $\nu = 0.49$	
Fat		$E = 1 \sim 10 \text{ kPa}$, $\nu = 0.49$	
Breast	Linear Elasticity	$E = 330 \text{ kPa}$, $\nu = 0.21$	
Skin	Linear Elasticity	$E = 500 \text{ kPa}$	Liu et al. (2023)
Pectoralis major muscle	Linear Elasticity	$E = 10 \text{ kPa}$	Chen et al. (2024)
Glandular tissue	Linear Elasticity	$E = 10 \text{ kPa}$	
Ligaments	Linear Elasticity	$E = 100 \text{ kPa}$	
adipose tissue	Mooney-Rivlin	$C_{10} = 0.31 \text{ kPa}$, $C_{01} = 0.3 \text{ kPa}$, $C_{11} = 2.25 \text{ kPa}$, $C_{20} = 3.8 \text{ kPa}$, and $C_{02} = 4.72 \text{ kPa}$	

from supports, ensuring that the analysis results are both accurate and reliable. This section primarily reviews breast deformation under five different conditions, as illustrated in Figure 1.

2.2.1. Breast motion

Dynamic deformation of the breast during movement can lead to discomfort, negatively impacting physical activities (Gilmer et al. 2024). Chen et al. (2013) utilized a three-dimensional dynamic finite element model to simulate nonlinear breast motion, effectively modeling and predicting breast movement under conditions such as physical activities. They constructed a model comprising the thoracic cavity, two breasts, and three layers of skin. By using a motion capture system to record the coordinates of subjects' breasts while walking, jogging, and ascending or descending stairs, the accuracy of their simulations were thereby validated with experimental data. Cai et al. (2018) developed a theoretical mathematical model to investigate the viscoelastic behavior of the breast during free vibration and movement, based on the breast vibration waveforms obtained from free-fall experiments, and established a 3D finite element model that included the torso and breast to predict breast motion. Zhang et al. (2022) proposed a nonlinear biomechanical finite element model that incorporates complex anatomical structures, such as the torso, breast, pectoralis major, and skeleton, allowing for the simulation of soft tissue deformation during arm abduction from 30° to 90°. They collected breast deformation data through a motion capture system to validate the finite element model. Chen et al. (2024) employed 4D scanning technology and a motion capture system to collect experimental data, constructing a geometric model that includes rigid bodies, soft tissue layers, skin, pectoralis major, fat, suspensory ligaments, and glandular tissue. They used a nonlinear multi-component dynamic finite element method to simulate breast motion, analyzing displacements, biomechanical factors, and the stiffness effects of various internal structural components of the breast through simulations of different constituent parts.

2.2.2. Bra constraint

Wearing a bra can significantly reduce vertical displacement of the breasts during movement, thereby decreasing the perception of breast discomfort (Leme et al. 2020). Compared to regular bras, sports bras are more effective in minimizing breast movement (Sheng et al. 2021), providing better support and comfort. The interaction between breasts and bra involves complex force transfer and nonlinear

problems, which the underlying biomechanical mechanism is not easy to reveal *via* traditional experiments. This underscores the importance of establishing a finite element contact model between the bra and the breast in studying breast-bra interaction. Sun et al. (2019) introduced an innovative personalized modeling system based on a finite element contact model for bras and the human body, designed to simulate the shaping effects of bras on breast morphology and their pressure distribution on the skin. They developed a finite element model based on individual-specific geometries to simulate the interaction between breast cups made of different material properties and the breasts (Sun et al. 2021). The findings indicated that stiffer fabrics performed better in reshaping breast morphology and reducing shoulder strap pressure on the neck, but resulted in higher pressure distribution on the lower part of the breasts. Liu et al. (2023) proposed a novel optimization method that utilizes CAD to simulate breast deformation during jumps while wearing a sports bra, integrating finite element assembly with three-dimensional scanning, reverse engineering, and boundary conditions. Liang et al. (2019) conducted finite element analyses to simulate both static and dynamic friction conditions as well as dynamic displacements of the breasts, employing motion capture data to measure damping ratios and calculate the natural frequencies of exposed breasts and breasts supported by sports bras.

2.2.3. Compressive deformation

Mammography, a common medical imaging technique, is particularly utilized for breast cancer screening and diagnosis. Finite element analysis can simulate the compressive deformation of the breast during mammographic procedures, aiding in the optimization of image quality and diagnostic accuracy, as well as simulating the compression process occurring during MRI-guided biopsies. Azar et al. (2000) proposed a novel method for guiding clinical breast biopsies based on a variable deformable finite element model, allowing for imaging of the breast in an uncompressed or mildly compressed state prior to surgery, followed by compression of the breast to predict the tumor location during the procedure using the finite element model. Chung et al. (2008) developed a biomechanical model to simulate breast compression during mammographic imaging, applying 32% compression to a patient's breast using a compression plate within an MRI coil, followed by MRI imaging. They employed finite deformation elasticity theory, contact mechanics, and individualized tissue

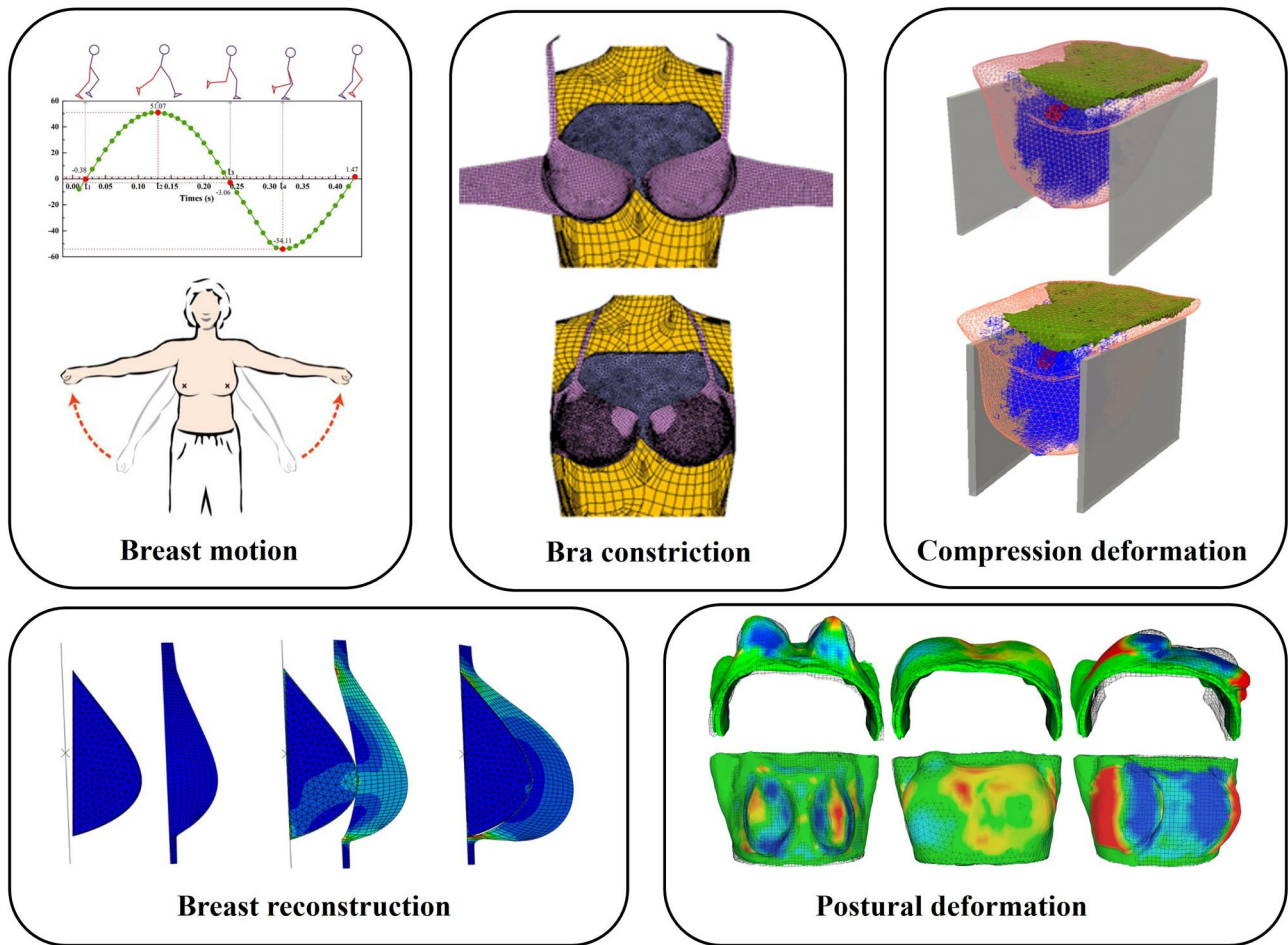


Figure 1. Five patterns of breast deformation.

characteristics to numerically simulate the finite element model of the left breast. Sturgeon et al. (2016) utilized finite element methods to develop a series of computational breast models based on CT data, creating a program that allows users to modify the model to simulate the effects of gravity and breast compression, making the model suitable for multimodal imaging. Han et al. (2012) introduced a breast biomechanical modeling framework for patients, simulating and analyzing a series of five breast compression experiments consisting of supporting structures, fixed plates, and movable plates to compress the breast, resembling the compression process during mammography or MRI-guided biopsies.

2.2.4. Deformation simulation in different positions

There are notable differences in breast positioning during medical imaging and surgical procedures, which can be predicted through finite element analysis. del Palomar et al. (2008) explored the correlation between the spatial characteristics of the breast in supine and standing positions, proposing a biomechanical model of the breast implemented in a finite element environment

to predict breast deformation in different positions, comparing the predictions with multimodal images. Míra et al. (2018) developed a novel biomechanical breast model using finite element theory, calculating the geometric shapes of three breast configurations—prone, supine, and supine inclined—based on stress-free geometric measurements and estimated equivalent Young's moduli to assess the deformation between estimated and measured breast geometries. Chen et al. (2019) employed a computer algorithm-based automatic segmentation method to segment the entire breast and fibro-glandular tissue on magnetic resonance imaging (MRI), applying a gravitational loading transformation and utilizing a finite element model to deform prone MRI images to align with supine ultrasound images. Mazier and Bordas (2024) proposed a simulation pipeline to compute patient-specific geometric shapes and predict individualized breast material properties, calibrating the geometric material model through calculations of the undeformed breast state and identifying the mechanical characteristics specific to the patient's breast and skin to obtain optimal estimates for the supine position.

2.2.5. Deformation simulation in breast reconstruction

Breast reconstruction surgery is a crucial aspect of post-operative recovery for breast cancer patients, aimed at restoring the shape, size, and symmetry of the breast. While traditional surgical techniques can achieve certain reconstructive outcomes, it is still a challenge to precisely control the deformation of breast tissue. The finite element method (FEM) has become a key tool in breast reconstruction research due to its ability to handle complex material properties and geometries. Roose et al. (2005) developed a computational model to simulate the breast shape of a 30-year-old woman after the implantation of a 245-gram silicone-filled implant. They compared a mass-spring system with a finite element model and investigated the effects of different elastic models. Thanoon et al. (2015) employed a finite element model for virtual surgery, studying the breast models of three women with breast cancer. This study conducted sensitivity analyses on excision location, excision size, mechanical properties of breast tissue, and different postoperative recovery stages to gain preliminary insights into key parameters that could yield satisfactory cosmetic results after breast-conserving surgery. Haddad et al. (2016) utilized a nonlinear finite element model to simulate breast reconstruction following mastectomy and lumpectomy, assessing the deformation of reconstructed breasts using fat grafts under various loading conditions in both prone and upright positions. Dong et al. (2016) used MR images of patients' breasts to propose a patient-specific finite element approach, investigating the effects of two similarly shaped but differently sized implants on augmentation outcomes, as well as changes in breast morphology in supine and standing positions. Myung et al. (2019) employed computational finite element analysis to explore changes in morphology and stress distribution following breast implantation, performing static analyses of implant insertion locations and dynamic analyses of downward pressure exerted by gravitational forces during physical activities.

2.3. Validation and accuracy assessment of finite element models

The aforementioned studies employ the finite element method to simulate breast deformation across various physiological and clinical scenarios. To ensure the reliability of these simulations, model validation and testing are of paramount importance. The core of validation lies in the quantitative comparison of finite element predictions with independent clinical measurements or experimental data.

Common validation methods include: (1) Comparison with medical imaging data: For instance, performing three-dimensional registration and comparison between the simulated compressed breast morphology and the actual morphology under mammography or magnetic resonance imaging (MRI) (Chung et al. 2008; Han et al. 2012); or comparing the predicted supine deformation with the patient's actual supine ultrasound images (Chen et al. 2019). (2) Comparison with dynamic experimental data: Using motion capture systems (Chen et al. 2013; Zhang et al. 2022; Chen et al. 2024) or 4D scanning technology (Chen et al. 2024) to record the true displacement trajectories of skin markers during human activity, which serve as the gold standard for assessing the motion accuracy of corresponding nodes in the simulation.

To quantify predictive accuracy, studies employ several metrics: Point-to-point distance error (e.g. the average Euclidean distance between predicted nodal positions and measured positions) serves as a direct measure (Zhang et al. 2022; Chen et al. 2024); surface distance error (e.g. the average symmetric surface distance) is used to evaluate overall shape similarity (del Palomar et al. 2008; Míra et al. 2018); and volume overlap (e.g. the Dice coefficient) assesses the accuracy of internal tissue (e.g. glandular tissue) deformation prediction (Chen et al. 2019). For example, Chen et al. (2024) reported that the average error between their multi-component model's prediction of nipple displacement and motion capture data was less than 3 mm; del Palomar et al. (2008) reported an average error of approximately 2.6 mm between their model's predicted breast surface point positions and MRI measurements. These quantitative metrics provide objective evidence for the credibility of the models within specific application contexts.

2.4. Advantages of the finite element method and prospects for integration

The Finite Element Method (FEM) has established itself as a benchmark tool for the biomechanical simulation of breast soft tissues, owing to its unique capabilities in modeling complex geometries and nonlinear material behavior. The breast exhibits a highly individualized, asymmetric, and complex curvilinear geometry, incorporating intricate anatomical details such as skin folds and interwoven glandular and adipose tissues. By discretizing the continuum into a mesh, FEM supports diverse element types and adaptive mesh refinement, enabling the precise capture of

these complex morphological features. Unlike the Finite Difference Method, which is constrained by regular grids, FEM naturally conforms to irregular computational domains. Furthermore, compared to mass-spring systems, FEM is rooted in continuum mechanics. The use of element shape functions ensures the continuity of field variables, maintains volume conservation, and allows for the accurate computation of internal, continuous stress and strain fields. Additionally, its mature contact algorithms effectively address complex sliding contact problems between the breast and external surfaces, such as bras or compression plates.

Breast tissue demonstrates marked hyperelasticity, viscoelasticity, near-incompressibility, and spatial heterogeneity. FEM can integrate hyper elastic constitutive models—such as Mooney-Rivlin and Neo-Hookean—to accurately describe nonlinear mechanical responses under large deformations *via* strain energy density functions. It also facilitates the straightforward assignment of heterogeneous material properties to different regions of the breast, thereby enabling more realistic simulations of tissue mechanics. Moreover, FEM allows for the convenient application of displacement constraints, pressure loads, or spatially varying complex loading conditions, making it ideally suited for simulating a wide range of practical scenarios such as motion and compression.

Although traditional FEM is accurate, its high computational cost hinders real-time application. Integrating FEM with machine learning presents a promising pathway to enhance efficiency and achieve real-time performance, potentially reducing prediction times from several hours to milliseconds. This advancement could unlock clinical applications such as intraoperative navigation. Crucially, FEM provides physics-based constraints. Incorporating physical governing equations as regularization terms into the loss function of a machine learning model guides the model to learn the inherent data relationships that comply with physical laws. Consequently, even in regions outside the training data distribution, the model can yield more physically plausible predictions, thereby enhancing its generalizability and robustness.

3. Applications of machine learning in breast deformation analysis

Machine learning has a wide range of applications in the field of breast medicine, primarily including early detection and diagnosis of breast cancer (Uhlig et al. 2018; Fatima et al. 2020; Mahmood et al. 2020; Chugh et al. 2021; Houssein et al. 2021; Bacha and Taouali

2022; Mahmood et al. 2024; 2025), morphological prediction and optimization in breast reconstruction surgery (Wang et al. 2019; O'Neill et al. 2020; Basile and Oliveira 2024), postoperative outcome assessment and personalized treatment recommendations (Pfob et al. 2021; Naoum et al. 2022; Braun et al. 2023; Chen et al. 2024; Oleck et al. 2024), as well as the integration and standardized management of breast medical data (Pfob et al. 2022; Cho et al. 2023). These applications not only enhance diagnostic and therapeutic efficiency as well as accuracy, but also provide patients with more precise and personalized medical services. In addition to conventional machine learning approaches, recent advances in deep learning have further enhanced breast deformation analysis, particularly under dynamic conditions. Dynamic breast motion during compression, postural changes, or locomotion involves strong nonlinear and spatio-temporal coupling that is difficult to capture using static or simplified models. Deep learning architectures, such as convolutional neural networks and temporal sequence models, are capable of learning complex relationships between anatomical features, loading conditions, and deformation responses directly from medical imaging data and FEM-generated simulations. When trained on FEM datasets, these models can act as efficient surrogate solvers, enabling rapid prediction of displacement and strain fields while preserving key biomechanical characteristics (Kirchdoerfer and Ortiz 2016; Bock et al. 2019). Beyond supervised learning, emerging paradigms such as reinforcement learning and unsupervised learning provide additional perspectives for breast deformation analysis. Reinforcement learning may be applied to optimize biomechanical or clinical objectives, such as identifying breast compression strategies in imaging that balance image quality and patient comfort. Unsupervised learning techniques (Mahmood et al. 2024) can extract latent deformation patterns and reduce the dimensionality of high-resolution FEM outputs, which is particularly valuable when labeled clinical data are limited. When combined with physical constraints or governing equations, data-driven models can maintain physical consistency and robustness (Raissi et al. 2019).

3.1. Data acquisition and collection

In the analysis of breast tissue deformation, the development of machine learning models begins with the acquisition and integration of multi-source data. The use of patient-specific data must adhere to strict ethical guidelines and comply with data-protection regulations such as the GDPR and China's Personal Information

Protection Law. Prior to any processing, data are anonymized through the removal of all personal identifiers. To further preserve privacy in distributed or multi-institutional settings, privacy-enhancing techniques such as federated learning are considered; these allow collaborative model training by exchanging only parameter updates without sharing raw patient data, thereby supporting secure, compliant multicenter research. The data sources are primarily divided into two categories: first, medical imaging data itself, which provides the most direct anatomical and morphological input for the models; and second, associated clinical and biomechanical information. Prior to being input into models, imaging data (such as X-ray (Wels et al. 2012; Zolfagharnasab et al. 2018; Fanizzi et al. 2020; Rehman et al. 2025; Mahmood et al. 2022; Mahmood et al. 2025), ultrasound (Dołęga-Kozierowski et al. 2023), and MRI (Zolfagharnasab et al. 2018; Mendizabal et al. 2020; Said et al. 2022; Mahmood et al. 2024)) must undergo rigorous preprocessing, including but not limited to resampling to a uniform resolution, spatial normalization to align anatomical coordinate systems across different patients, and necessary denoising or enhancement procedures. For dynamic analysis, temporal registration of the sequences is also required.

Simultaneously, clinical information (such as patient age, BMI, tumor location, and staging) and quantitative features derived from imaging (such as glandular density, tissue volume, and texture features) are structured into tabular data or low-dimensional vectors. These multimodal data can be integrated through feature concatenation, early fusion, or attention-based mechanisms, collectively forming the input for machine learning models. This enables the establishment of a mapping relationship from individualized features to complex deformations or clinical endpoints (Martínez-Martínez et al. 2017; Zolfagharnasab et al. 2018; Fanizzi et al. 2020).

Bacha and Taouali (2022) proposed a differential evolution algorithm for diagnosing breast cancer from mammographic images. Said et al. (2022) introduced a novel automatic method that utilizes unsupervised neural networks to simultaneously segment three tissue types in T2-weighted breast MRI. Kurrant et al. (2021) presented a robust and flexible medical image segmentation technique for classifying microwave breast images into tissue types to facilitate image quality assessment.

By employing datasets collected through various methods, machine learning models can be further trained and optimized, enhancing their precision and efficiency in handling specific tasks. Wels et al. (2012) developed a method for 3D breast decompression and

related lesion mapping from 3D digital breast tomosynthesis (DBT) data. This approach is entirely data-driven and uses machine learning techniques to predict the shape of the uncompressed breast derived from the DBT input volume. Martínez-Martínez et al. (2017) introduced a data-driven method that employs finite element analysis for offline simulation of breast deformation caused by two compression plates, utilizing three machine learning models to learn the relationship between inputs (geometry, breast tissue, and loads) and outputs (nodal displacements) in the simulations. Zolfagharnasab et al. (2018) proposed an internally semi-synthetic dataset to facilitate the prediction of breast deformation following breast-conserving surgery through machine learning techniques.

3.2. Feature selection and extraction

Feature extraction and representation are critical steps in machine learning tasks. Feature selection involves identifying and selecting the most relevant features that effectively convey key information about breast deformation, while feature extraction transforms the selected features into a format that can be processed by models. This process aids to reduce data dimensionality, enhance training efficiency, and minimize the risk of overfitting, potentially improving the model's generalization capabilities (Rehman et al. 2025). Features related to breast deformation can be extracted and analyzed from various perspectives, including density (Zolfagharnasab et al. 2018), shape (Wang et al. 2019), volume (Wels et al. 2012; Zolfagharnasab et al. 2018), laterality (Zolfagharnasab et al. 2018), displacement (Mendizabal et al. 2020; Zhang et al. 2022; Zhou et al. 2022), texture (Moore et al. 2010; Fanizzi et al. 2020), and implants (Basile and Oliveira 2024). The key to feature selection lies in choosing those that most effectively characterize the structural and morphological changes in the breast, as these features will serve as the basis for training and predicting with the machine learning models.

Zhang et al. (2022) filtered 14 variables from subjects using gray relational analysis to identify 8 factors that significantly impact breast displacement, which were then used as input features to predict the three-dimensional deformation of aging breasts. Zolfagharnasab et al. (2018) considered breast shape, volume, density, and tumor characteristics as inputs for the machine learning model to predict breast deformation following breast-conserving surgery. Zhou et al. (2022) utilized a motion capture system to collect displacement data from 10 markers placed on the breasts of 11 elderly

women during yoga practice, which served as features for predicting age-related breast deformation. Fanizzi et al. (2020) proposed a fully automated model for extracting breast texture features, primarily based on texture analysis methods, to characterize clustered microcalcification areas in digital mammograms.

3.3. Model selection and training

Model selection should be based on data characteristics, the intended analysis objectives, and computational resources. Commonly used machine learning models in breast-related research include decision trees (Martínez-Martínez et al. 2017), random forests (Wels et al. 2012; Zolfagharnasab et al. 2018), gradient boosting regression (Zolfagharnasab et al. 2018; Pfob et al. 2021; Cho et al. 2023), and neural networks (Mendizabal et al. 2020; Pfob et al. 2021; Said et al. 2022; Zhang et al. 2022). Diagrams of decision trees and neural networks are shown in Figure 2 and 3. Model training is an iterative process that requires the use of labeled datasets for supervised learning. By employing optimization algorithms, model parameters and structures are adjusted to enhance the accuracy and stability of the model in predicting breast deformation.

Islam et al. (2020) conducted a comparative study on the application of five machine learning techniques, i.e. support vector machines, K-nearest neighbors, random forests, artificial neural networks, and logistic regression, in breast cancer prediction. Uhlig et al. (2018) trained diagnostic models on clinical breast datasets using five machine learning techniques, including random forests, backpropagation neural networks (BPN), extreme learning machines, support vector machines, and K-nearest neighbors. Li and Chen (2018) employed five different classification models which are decision trees (DT), random forests (RF), support vector machines (SVM), neural networks (NN), and logistic regression (LR) to classify two different datasets related to breast cancer. Martínez-Martínez et al. (2017) trained a decision tree and two tree-based ensemble methods to estimate the deformation experienced by the breast under compression in real time. Wang et al. (2019) utilized an unsupervised K-means clustering method to categorize breast types and constructed corresponding implants for each type. Wels et al. (2012) applied machine learning techniques to predict the shape of an uncompressed breast from digital breast tomosynthesis (DBT) input volumes, employing multivariate random forest regression to forecast shape parameters. Zolfagharnasab et al. (2018) used random forests, gradient boosting regression, and multi-output regression models to predict

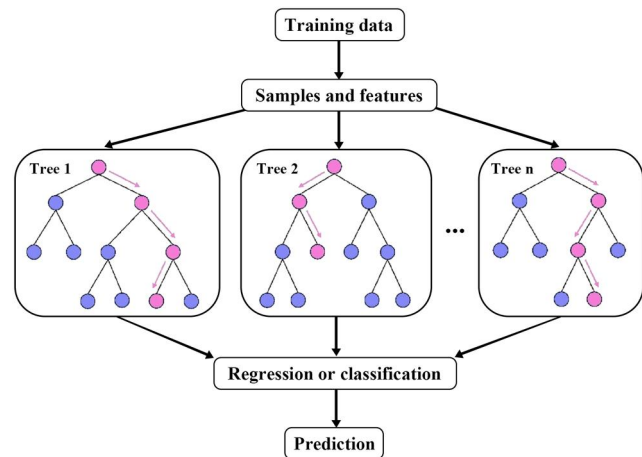


Figure 2. Schematic diagram of DT.

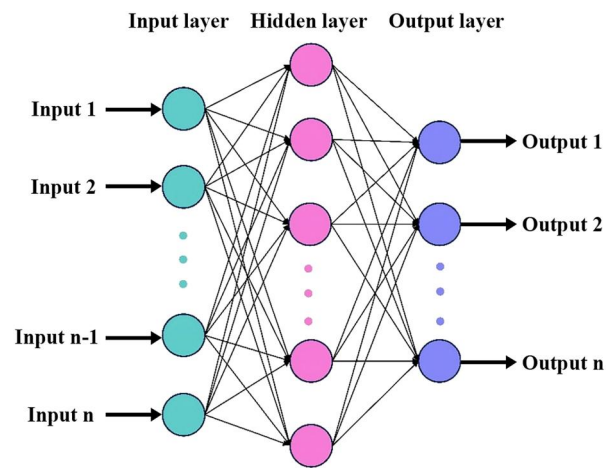


Figure 3. Schematic diagram of NN.

breast deformation after breast-conserving surgery. Zhang et al. (2022) applied particle swarm optimization with backpropagation artificial neural networks to predict real-time strain on breast skin during physical activity in older adults. Pfob et al. (2021) trained logistic regression (LR) with elastic network penalties, extreme gradient boosting (XGBoost) trees, and neural networks to predict patient-reported outcomes in women undergoing cancer-related breast surgeries and reconstructions during a two-year follow-up.

3.4. Model validation and testing

Model validation and testing are essential steps in evaluating the performance of machine learning models. By assessing models on independent test datasets, researchers can verify their accuracy, reliability, and generalization capabilities. Test results can inform further adjustments to model parameters, and cross-validation techniques (Pfob et al. 2021; 2022; Chen et al. 2024) can be employed to optimize model training and parameter

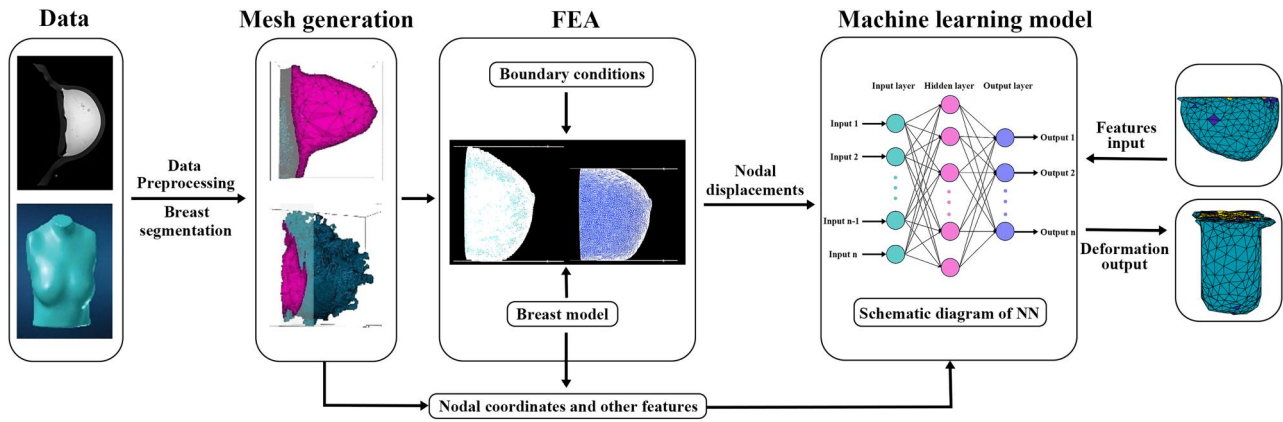


Figure 4. Flowchart of the finite metadata-driven machine learning model approach.

selection, or to modify the model architecture as needed. In breast deformation analysis, displacement error (Wels et al. 2012; Martínez-Martínez et al. 2017; Mendizabal et al. 2020; Zhang et al. 2022) is commonly used as an indicator of machine learning model performance.

In summary, machine learning technologies demonstrate tremendous potential in breast medical imaging and disease diagnosis. Particularly in breast deformation analysis, these technologies can effectively achieve real-time deformation predictions, providing significant support and innovation for medical diagnosis. Future research may further optimize data acquisition and processing methods, explore more complex deep learning models to enhance the accuracy and generalizability of deformation analysis. Additionally, it is crucial to align with clinical needs, considering medical knowledge and individual patient differences to develop personalized breast deformation diagnostic and management strategies, ultimately leading to better medical outcomes and treatment plans.

4. Application of machine learning models driven by finite element data in breast deformation

The integration of finite element modeling and machine learning has direct clinical relevance in breast cancer diagnosis, personalized treatment planning, and breast-related medical device design. During mammography and digital breast tomosynthesis, breast compression induces substantial tissue deformation, leading to discrepancies between observed and true lesion locations (Samani et al. 2003). Machine learning models trained on FEM-simulated deformation fields can rapidly estimate patient-specific three-dimensional deformation, thereby improving tumor localization accuracy and biopsy guidance (Kirchdoerfer and Ortiz 2016). In surgical planning and breast reconstruction, FEM–ML

frameworks enable fast evaluation of postoperative breast shape and internal stress distributions for different treatment scenarios, supporting more objective and personalized clinical decision-making (Han et al. 2012). From an engineering perspective, FEM–ML integration also supports efficient optimization of breast-related medical devices, including compression plates, implants, imaging coils, and supportive garments. FEM provides detailed insight into tissue–device interactions, while machine learning accelerates parameter exploration through surrogate modeling, reducing computational cost and facilitating patient-specific customization.

4.1. Data-Driven methods

Finite element data-driven machine learning methods involve the application of machine learning techniques to finite element analysis (FEA) data to enhance predictive accuracy while reducing computational costs and time, as illustrated in Figure 4. This approach is particularly suitable when traditional finite element analysis becomes impractical or unfeasible due to the complexity of the model, limitations in computational resources, or time constraints. It offers a new pathway for the rapid and efficient simulation of breast tissue deformation. By integrating the physical accuracy of FEA with the predictive speed of machine learning, these hybrid models hold significant potential for clinical applications.

Previous studies have proposed machine learning methods based on the Finite Element Method (FEM) to predict deformations in biological models, including the liver (Abdelrahman et al. 2011; Lorente et al. 2017), stomach (De et al. 2011), and arteries (Madani et al. 2019). Martín-Guerrero et al. (2016) introduced a machine learning framework for learning FEM modeling, conducting a feasibility analysis for modeling the biomechanical behavior of the liver and breast.

Martínez-Martínez et al. (2017) presented a data-driven approach to simulate breast compression, employing three machine learning models trained on data from these simulations to predict breast tissue deformation in real-time during compression. Liang et al. (2021) developed a finite element contact model between the human body and a bra, applying validation results to train a neural network model for predicting breast displacement based on various stretch moduli of bra straps. Wang and Kesavadas (2022) utilized finite element simulation results of breast compression to construct a neural network model, creating a surgical simulator to predict breast deformation under different material structures and patient positions. In another study, Wang and Kesavadas (2022) conducted finite element analysis on deformation data of multiple breasts in different positions to train a neural network for predicting surface deformation and tumor centroid locations. Said et al. (2023) employed machine learning algorithms to simulate the mechanical response of breast tissue under gravitational loading and during mammographic compression.

4.2. FEA simulation and dataset generation

Martin-Guerrero et al. (2016) selected the Neo-Hookean hyperelastic model to represent the biomechanical behavior of breast tissue, using a breast anatomy simulator to generate ten virtual breast phantoms of identical volume. They performed finite element analysis to assess the deformation of breasts under compression, employing seven features to train three models. Martínez-Martínez et al. (2017) utilized the hyperelastic Mooney-Rivlin material model for glandular and adipose tissues proposed in previous studies (Yin et al. 2004), selecting ten real breast images for offline finite element simulations. From these simulations, 11 features were extracted to train three machine learning models for real-time prediction of breast tissue deformation under compression. Liang et al. (2021) employed five Mooney-Rivlin coefficients (Sun et al. 2019) to simulate the biomechanical behavior of the breast, establishing a three-dimensional contact model between the body and a sports bra using finite element software. This model simulated the running activity of subjects, allowing the construction of a neural network model that used the stretch modulus of the strap fabric as input features to determine the relationship between time and breast displacement during running. Wang and Kesavadas (2022) adopted the Neo-Hookean hyperelastic model as the material and implemented the material estimation method proposed by Visentin

et al. (2019), utilizing finite element analysis to simulate the mechanical behavior of the breast in different deformation states. They collected node displacement data during the simulations to train an advanced neural network, which took the direction of gravitational force or the ratio of glandular tissue as input parameters to predict the displacement vectors of surface nodes and the coordinates of tumor centers. Said et al. (2023) normalized 516 breasts and employed an isotropic hyperelastic Neo-Hookean material model to simulate breast compression, predicting deformation based on a biomechanical model derived from breast MRI under actual mammographic compression.

In addition to these studies, dynamic experiments have provided rich data resources for the biomechanical research of breast tissue. For instance, deformation data of the breast during dynamic postures such as arm elevation (Zhang et al. 2022) and yoga (Zhou et al. 2022), along with finite element simulations of breast tissue deformation and stress distribution under load (Martínez-Martínez et al. 2017), further enrich the dataset on breast tissue deformation. This comprehensive dataset enhances the predictive capability of machine learning models regarding breast tissue behavior.

4.3. Model selection and training

The datasets for training machine learning models primarily originate from finite element analysis of breast deformation, encompassing breast characteristics, finite element mesh features, node coordinates, and other attributes, as detailed in Table 2. Martínez-Martínez et al. (2017) compared three machine learning models (decision trees, random forests and extremely randomized trees), optimizing hyperparameters to predict breast compression deformation. Liang et al. (2021) utilized finite element simulation data to train a neural network model to predict the relationship between breast displacement and time during running. Wang and Kesavadas (2022) employed neural network models to predict breast deformation under different postures and to estimate tumor centroid locations. Said et al. (2023) applied extremely randomized trees (ERT), extreme gradient boosting (XGBoost), and a recurrent neural network (RNN) model to predict breast deformation in both unloaded and compressed states. These studies not only demonstrate the potential applications of various machine learning algorithms in predicting breast deformation but also emphasize the critical roles of data preprocessing, model selection, and hyperparameter tuning in enhancing predictive accuracy.

Table 2. Literature summary of FE-driven machine learning models in breast deformation.

References	Material type of breast tissue	Number of breasts	Breast deformation	Machine learning model	Input features	Output
Martin-Guerrero et al. (2016)	Neo-Hookean	10	Breast compression	DT RET RF	1. The three nodal Euclidean coordinates from each breast mesh 2. The percentage of glandular tissue of each breast 3. The number of fat and glandular elements from the mesh to which each node belongs 4. The value of the plate displacement in each step	Nodal displacements
Martínez-Martínez et al. (2017)	Mooney—Rivlin	10	Breast compression	DT ERTs RF	5. The three nodal Euclidean coordinates from each breast mesh 6. The fraction of skin, fat, and glandular tissue of each breast 7. The normalized volume of each breast 8. The normalized number of skin, fat, and glandular elements from the mesh to which each node belongs 9. The value of the plate displacement in each step	Nodal displacements
Liang et al. (2021)	Mooney—Rivlin	1	Running deformation under bra constriction	NN	The tensile modulus of the shoulder strap fabric	Nodal displacements
Wang and Kesavadas (2022)	Neo-Hookean	–	Gravitational deformation in different posture	ANN	1. The vector of R^4 2. The gravity direction 3. The gland ratio	1. Nodal displacements 2. The tumor location
Wang and Kesavadas (2022)	Mooney—Rivlin	–	Gravitational deformation in different posture	ANN	1. The vector of R^4 2. The gravity direction 3. The gland ratio	1. Nodal displacements 2. The tumor location
Said et al. (2023)	Neo-Hookean	516	Breast compression	ERT XGBoost RNN	1. The geometry of the breast described by the initial coordinates of the breast mesh 2. The properties of the tissues such as the breast volume and the volume fractions of the tissues given by the segmentation 3. The external factors causing the deformation such as the position of the compression plates and surface nodes of the breast mesh	Nodal displacements

4.4. Model validation and application

The dataset derived from finite element analysis (FEA) simulations is typically partitioned to facilitate the development and rigorous evaluation of data-driven models. In machine learning practice, especially when dealing with complex models like neural networks, the data is commonly divided into training, validation, and testing subsets. The training set is used to fit the model parameters. The validation set serves as an independent benchmark for tuning hyperparameters, selecting among model architectures, and preventing overfitting during the training process. Finally, the held-out testing set provides an unbiased assessment of the model's generalization performance on completely unseen data, which is the ultimate measure of its practical utility. This three-way split is considered a best practice for robust model development, as exemplified in studies such as that by Liang et al. (2021), who employed a 70:15:15 ratio for training, validation, and testing, respectively.

In machine learning research based on finite element data, model accuracy is usually validated by comparing model predictions with finite element simulation results. Martínez-Martínez et al. (2017) utilized two validation methods, calculating the average distance between the predicted node positions and those extracted from finite element simulations to assess model performance on the validation set. The extreme randomized tree model outperformed the other two models, with all three models achieving an average error of less than 2 mm in predicting node displacements, which aligns with the clinical threshold set for applications. Wang and Kesavadas (2022) allocated 20% of the data for validation and 80% for training; the results showed that the average point validation NDE (normalized distance error) for the breast was less than 0.014 mm, while the NDE for the tumor center was below 0.05 mm. This model can provide real-time predictions of breast deformation during treatment and can also be utilized to create tools for accurate tumor localization in the operating room. Said et al. (2023) divided their dataset into 90% for training and 10% for evaluation, employing the root mean square error (RMSE) as the loss function. They assessed the model's accuracy in relation to clinical dataset features, such as compression ratio, breast volume, and tissue type. The average RMSE estimates for unloaded states and compression simulations were 3.4 and 4.7 mm, respectively.

4.5. Challenges and prospects of technological integration

The integration of the Finite Element Method (FEM) and Machine Learning (ML) offers an innovative

framework for the precise analysis of breast deformation. However, its effective deployment in clinical settings faces several interconnected challenges spanning three key dimensions: data, model reliability, and computational efficiency, all of which impact real-time clinical feasibility.

A primary bottleneck is the scarcity of high-quality, patient-specific simulation data. Building reliable individualized models requires extensive FEM data derived from medical imaging, yet the high cost and complexity of acquisition often result in limited and non-diverse datasets. This data sparsity and heterogeneity promote model overfitting and degrade performance for cases underrepresented in training (e.g. specific body types or complex pathologies), limiting clinical generalizability. Furthermore, the 'black-box' nature of many ML models undermines clinical trust, and purely data-driven predictions risk violating fundamental physical laws, affecting their acceptability for safety-critical decisions. The combined computational burden of high-fidelity FEM simulation and complex ML model training also poses a significant barrier, making it difficult to meet the millisecond-level feedback required for applications like surgical navigation.

Research is advancing multi-layered solutions to these challenges. To address data scarcity, Generative Adversarial Networks (GANs) synthesize physiologically plausible training data, while transfer learning leverages external domain knowledge to enhance generalization on small datasets. To improve model credibility, Physics-Informed Machine Learning embeds physical constraints (e.g. governing equations) directly into the learning process, ensuring predictions adhere to known biomechanical laws. For computational efficiency, the development of lightweight surrogate models is key. High-fidelity finite element models and deep learning models are typically associated with high computational costs. Model order reduction and surrogate modeling techniques can significantly reduce computational complexity while preserving essential mechanical characteristics (Kirchdoerfer and Ortiz 2016; Bock et al. 2019). Meanwhile, techniques such as model pruning (Han et al. 2015), parameter quantization (Hubara et al. 2018), and knowledge distillation (Hinton et al. 2015) are widely employed to compress and accelerate neural network inference. In addition, specialized architectures such as graph neural networks, which are naturally suited to finite element mesh data, can further enhance computational efficiency while maintaining physical consistency. Furthermore, physics-informed neural networks

explicitly incorporate mechanical constraints into the learning process, ensuring the physical plausibility of predictions while improving computational efficiency (Raissi et al. 2019), thereby substantially enhancing the feasibility of FEM–ML approaches for real-time or near real-time clinical applications.

The integration of FEM and ML presents a novel pathway with significant clinical potential. In breast cancer diagnosis, it can rapidly predict 3D deformation under compression, enabling more accurate mapping of lesions from 2D images to a patient's 3D anatomy for improved biopsy targeting. For personalized surgical planning, it allows preoperative simulation of different tumor resection and reconstructive strategies, helping surgeons balance oncological safety with aesthetic outcomes. The technology also aids in predicting the long-term biomechanical behavior of breast implants to inform material and technique selection. Furthermore, in precision radiotherapy and rehabilitative device design (e.g. supportive bras), it can model tissue deformation due to positional changes or optimize soft tissue-textile interactions, providing an engineering basis for personalized treatment plans and product enhancements.

5. Conclusion

Finite element methods and machine learning in breast deformation analysis exhibit significant prospects and development potential. Finite element methods provide detailed predictions and optimization solutions for breast deformation through precise modeling and simulation, effectively capturing complex biomechanical features while also enabling practical solutions in clinical settings. Meanwhile, machine learning, as a data-driven approach, allows for rapid and efficient deformation prediction, particularly excelling in handling large-scale datasets. The combination of finite element methods and machine learning, resulting in data-driven finite element models, can significantly enhance predictive accuracy and greatly reduce the time required for deformation predictions. This integration not only optimizes the detail of physical modeling but also leverages the advantages of machine learning in analyzing complex datasets, enabling more precise and efficient assessments of breast deformation in clinical and apparel fields.

Future research should focus on the following directions to advance the field. On the data and methodology front, it is necessary to further optimize frameworks for multimodal data acquisition, standardized processing, and efficient fusion. The exploration of more sophisticated deep learning models—such as

physics-informed neural networks and spatio-temporal graph networks—should be pursued to enhance model accuracy, generalizability, and interpretability. At the level of clinical translation, the integrated technology should be closely aligned with clinical needs and applied to specific scenarios.

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Author contributions

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